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3° Longitudinal Periodicity in the UNESCO World Heritage Corpus: Enrichment in Domed Monuments and Harmonic Cluster Asymmetry

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ABSTRACT

A 3° longitudinal periodicity is identified in the externally curated UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$). Three enrichment tests survive Bonferroni correction at $k = 3$: full corpus ($p_{\text{adj}} = 0.005$), domed monuments ($N = 90$, $p_{\text{adj}} = 0.019$), and harmonic cluster asymmetry ($p_{\text{adj}} = 0.142$); all are cross-checked by permutation null models. A global anchor sweep places the ∼35°E corridor in the top 2.8% of 36,000 trial anchors. The Wikidata Q180987 stupa corpus and OWTRAD route network independently reproduce the pattern. The pipeline is fully reproducible.

1 | Introduction

This study reports a statistically significant 3° longitudinal periodicity in the externally curated UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$). The analysis proceeds in three steps. First, a unit sweep without any assumed anchor identifies 3° as the dominant longitudinal period (§4.1). Second, a phase-envelope scan demonstrates the signal is concentrated at a specific phase rather than uniformly distributed (§4.2). Third, three binomial enrichment tests that survive Bonferroni correction at $k = 3$ quantify the excess: the full corpus ($p_{\text{adj}} = 0.005$), domed monuments ($p_{\text{adj}} = 0.019$), and harmonic cluster asymmetry ($p_{\text{adj}} = 0.142$). All three are cross-checked by permutation and bootstrap null models (§4.6). Deviations are computed from a 3° grid referenced to Mount Gerizim (35.269°E), a historically documented candidate meridian (§2.2).

To apply the enrichment tests, sites are labeled by their proximity to the nearest 3° grid node using a fixed degree threshold (Tier-A+: $\delta \leq 0.06°$, $\lesssim$ 6.7 km, $P_0 = 4\%$). These tier widths are presentational bin boundaries, not inferential thresholds. The primary inference uses unbinned circular statistics (§3.3).

A global anchor sweep confirms the result is stable across the ∼35°E corridor (§5.2). A tier-threshold sensitivity audit confirms that enrichment is significant across null rates 1%–10% (Appendix B).

Methodological safeguards. Prior longitude-alignment studies were criticised for investigator-assembled site lists (Williamson & Bellamy 1983, Freeman 1976, Kendall 1974). This study differs: (1) the UNESCO corpus is externally curated and fixed; (2) the 3° period is identified without specifying an anchor (§6.2); (3) results are fully reproducible (Tenenbaum 2026). Because all inputs are fully public, every pipeline step is exploratory; no confirmatory inference is claimed.

2 | Background

2.1 | Prior Work on Spatial Periodicity and Sacred Landscapes

Statistical evaluation of spatial periodicity in monument distributions has a long and contested history. Thom's "megalithic yard" (Thom 1967 1971 1964) stimulated statistical refinement

(Kendall 1974) and methodological critique (Freeman 1976). Magli (Magli 2013 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles 1999 2015, Silva 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties.

2.2 | Reference Phase and Candidate Anchors

The harmonic deviations in this study are computed from Mount Gerizim (35.269°E) as the *a priori* reference phase. Gerizim is chosen because of its independently documented *tabbur ha-aretz* ("navel of the earth") cosmographic role in Samaritan tradition (Purvis 1968, MacDonald 1964, Crown 1989), a justification that predates the analysis entirely.

Caveat on anchor identification. The data do not uniquely identify an origin meridian. The von Mises mixture model (§3.4) recovers a phase offset relative to any fixed reference frame, but the fitted phase is an estimate subject to sampling uncertainty and cannot resolve which physical location defined the grid historically, or whether any single location did. Any UNESCO site that lies within the A+ tier of the Gerizim grid is, by construction, also a plausible reference point that would reconstruct nearly the same grid, because its fitted phase differs from Gerizim's by less than one tier width ($\lesssim$6.7 km). A site-as-anchor ranking audit (Supplementary Audit S6) confirms this. The top-ranked competing anchors, including the Silk Roads: Zarafshan-Karakum Corridor (62.21°E, deviation 0.061° from Gerizim harmonic $n = 9$), the Birthplace of Jesus (35.21°E), and Old City of Sana'a (44.21°E), all lie within the A+ tier of the Gerizim grid and therefore reconstruct an essentially identical periodicity. These are co-grid points, not independent alternatives.

The primary finding of this paper is therefore the existence of the periodicity itself; Mount Gerizim is presented as the historically motivated candidate for its phase origin, not as a conclusion the statistical analysis can establish independently.

A global anchor sweep (§5.2) confirms the result is not sensitive to the exact coordinate: every longitude in the x.18°E band produces the same enrichment, and both Gerizim and Jerusalem (35.232°E, ≈4.1 km south) place the corridor in the global top 2.8% of 36,000 trial anchors. Mecca ($p \approx 0.0571$, †), Mt. Meru ($p \approx 0.3051$, ns), and Mt. Kailash ($p \approx 0.2059$, ns) yield no significant enrichment, confirming the periodicity is phase-specific, even if the phase origin cannot be uniquely attributed.

Gerizim's coordinate is from the UNESCO tentative list (ref. 5706, DMS E35 16 08, State of Palestine, 2012; https://whc.unesco.org/en/tentativelists/5706/) and is not inscribed. Jerusalem is an inscribed corpus member, self-excluded when used as anchor.

3 | Data and Methods

3.1 | Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i–vi), *Natural* (criteria vii–x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (−235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 | Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E (see §2.2). Gerizim is not inscribed and does not appear in the standard corpus. Jerusalem is inscribed and scored as a regular site in Tests 1–4 (A+ at 4.1 km under Gerizim). For sensitivity sweeps (§5.2), Gerizim is added as a synthetic entry so each anchor self-excludes symmetrically.

For a site at longitude λ, the deviation from the nearest 3° harmonic is $\delta = |\lambda - 35.269| \bmod 3°$ (folded to $[0, 1.5°]$). The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim$ 6.7 km for Tier-A+) are equatorial approximations computed as $\delta \times 111$ km/°, becoming smaller at higher latitudes; the angular criterion is consequently conservative at high latitudes. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.06°/1.5° = 4\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.4) that preserve the empirical longitude distribution. A Gaussian KDE produces 40.4±6.2 expected A+ (~4%), confirming the geometric null is not materially inflated by longitude clumping.

3.3 | Statistical Framework

The primary inferential claims rest on three threshold-free tests (Tests 1–3, §3.6) that do not require any bin boundary or tier cutoff. All three survive Holm step-down correction at $k = 3$ and are described fully in §3.5.

Test 1 fits a two-component von Mises mixture to the dome subset (90 keyword-matched sites): one component fixed at the Gerizim phase on the 3° circle, one uniform background. The test statistic is the likelihood-ratio departure from a uniform null; significance is assessed by a $B = 10,000$ parametric bootstrap. No tier threshold is required; the Gerizim phase is a pre-specified geographic hypothesis, not a free parameter.

Test 2 computes the Spearman rank correlation between each harmonic node's site deviation and its cluster size across the full corpus ($N = 1011$), with a permutation null. It tests whether nodes drawing more sites also show smaller mean harmonic deviations — a structural signature of 3° periodicity that requires no tier threshold and no anchor.

Test 3 computes the mean harmonic deviation of the full corpus under a phase-rotation null that shifts the reference grid by a uniform random offset in $[0°, 3°)$. It tests whether the Gerizim phase specifically minimises the global mean deviation, without specifying any tier cutoff.

The Rayleigh test of uniformity (Mardia & Jupp 2000) on the 3° circle and the Rayleigh spectral periodogram (§4.6) are reported as supporting diagnostics. The raw Rayleigh averages uniformly over all sites and is structurally insensitive to a sparse (~9%) phase-concentrated sub-population; the periodogram confirms 3° is not a global longitudinal period, consistent with a signal that is concentrated at the Gerizim phase rather than omnidirectional.

Tier labels (A++, A+, A) defined in §3.4 are used as descriptive vocabulary to report harmonic proximity of individual sites; they do not enter any primary test statistic.

A von Mises mixture characterisation of two external corpora (OWTRAD and Wikidata Q180987 stupas) is reported in Appendix A as descriptive context only; it is not used to derive any inferential threshold.

3.4 | Tier Classification

Sites are assigned to tiers by their angular deviation δ from the nearest 3° harmonic node. Tier boundaries are fixed degree thresholds stored in `config.json`; their values were chosen to give geometrically convenient null rates (P_0 = 2%, 4%, 8% for A++, A+, A respectively). These are descriptive vocabulary for summarising harmonic proximity; no primary test statistic depends on a tier cutoff.

Tier A++ $\delta \leq 0.03°$ ($\lesssim 3.33$ km); $P_0 = 2\%$.

Tier A+ $\delta \leq 0.06°$ ($\lesssim 6.7$ km); $P_0 = 4\%$.

Tier A $\delta \leq 0.12°$ ($\lesssim 13$ km); $P_0 = 8\%$.

Tier B Middle zone, between harmonics and inter-harmonic midpoints.

Tier C/C−/C−− Symmetric mirror of A/A+/A++, measured from the nearest inter-harmonic midpoint.

3.5 | Test Populations

Keyword populations are approximate. The raw sweep tolerates known false positives while the context-validated variant (Test 2x) applies sentence-level gating for greater precision. Both populations are reported.

Test 1 (dome subset, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text to define the dome subset (N = 90). This raw keyword sweep is the sole inclusion criterion; no tier threshold is applied. The context-validated subset (N = 83) restricts to sites where the keyword appears in an architectural context; it is reported as a robustness check.

Test 2 (cluster asymmetry, primary). Full corpus (N = 1011), no keyword filter; tests whether harmonic nodes with smaller mean site deviation host more total sites (Spearman rank correlation), independent of any morphological keyword or tier boundary.

Test 3 (mean deviation, primary). Full corpus (N = 1011), no keyword filter; tests whether the observed mean harmonic deviation is smaller than expected under a phase-rotation null that shifts the reference grid uniformly in $[0°, 3°)$.

Test 2b (mound evolution, sensitivity variant). Extends the dome subset by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun*). Not counted separately in the Bonferroni family.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus based on year of inscription into the UNESCO corpus, no keyword filter.

Test 5 (founding classifier, post-hoc). A post-hoc keyword classifier identifies site types beyond domed monuments that fall on harmonic nodes; it carries no weight in the primary evidential chain (Appendix C.2).

3.6 | Multiple-Comparisons Protocol

Primary test family (k = 3). All three primary tests are threshold-free (no tier cutoff); significance is assessed by the Holm (1979) step-down procedure.

- **Test 1** (dome von Mises mixture, N = 90): two-component VM at Gerizim phase vs. uniform background; parametric bootstrap null (B = 10, 000). No tier threshold.
- **Test 2** (Spearman cluster asymmetry, full corpus, N = 1011): rank correlation between harmonic node deviation and site count; permutation null. No tier threshold.
- **Test 3** (mean deviation, full corpus, N = 1011): full-corpus mean harmonic deviation vs. phase-rotation null. No tier threshold.

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; a Mantel-Haenszel stratified test and logistic regression with region as a covariate assess the gradient robustly (§4.7); it is not part of the primary family. Test 2b is a sensitivity variant of Test 1 (dome subset). Test 5 (founding classifier) is post-hoc (Appendix C.2). World religion sub-group analyses (§4.3) use Fisher exact tests and are not Bonferroni-corrected. Tests A–D (§4.6) assess the 3° periodicity structure: Test A (full-corpus Rayleigh) is structurally insensitive to the sparse (∼3.1%) signal sub-population and is expected non-significant; Test D (targeted von Mises, full corpus) and the dome VM mixture (Test 1) together confirm phase concentration at the Gerizim direction; Test C confirms Gerizim is uniquely aligned relative to random anchors; Test B is tautological by construction and excluded from both tables.

3.7 | Simulation Null Models

Three simulation approaches (100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.4.

3.8 | Computational Reproducibility

All statistics are generated by an automated pipeline; every script writes to a shared results store and LaTeX macros are regenerated without manual editing. Scripts, audit files, and reproduction instructions are catalogued in Supplementary S1; see also §C.2).

4 | Results

The analysis proceeds in three evidential steps: (i) identification of 3° as the dominant longitudinal period without assuming any anchor (§4.1); (ii) phase localization showing the signal is concentrated at the Gerizim-grid phase rather than uniformly distributed (§4.2); and (iii) enrichment tests that quantify the excess against null models (§4.3–4.6).

4.1 | Periodicity Identification: Unit Sweep

Varying harmonic spacing from 1.5° to 6° (physical threshold fixed at 6.7 km), the 3° spacing produces the strongest joint significance across both the dome and full-corpus populations. The 1.5° and 6° spacings also reach significance, but both are mathematical consequences of the 3° structure: every 3° harmonic is also a 1.5° harmonic, and 6° harmonics are a strict subset, so neither constitutes an independent signal. Adjacent non-harmonic spacings (1.8°, 2.1°, 2.7°, 3.3°, 3.6°) are non-significant in either population, the key discriminating observation. The 2.4° spacing reaches nominal dome significance ($p = 0.0229$) but not full-corpus ($p = 0.4371$, ns), reflecting unsystematic local overlap with the 3° grid rather than a genuine secondary periodicity. Specifically, 6/9 (66.7%) of 2.4° dome hits align to the 3° grid, indicating the significance is due to overlap.

A degree-denominated Monte Carlo sweep (§6.2) confirms this pattern. Candidate intervals at 4° and 8° reach nominal dome significance, but inspection shows that 6 of 7 and all 4 of their hits respectively fall on 12° and 24° nodes shared with the 3° grid, confirming they are consequences of the primary periodicity rather than independent angular units.

A fine sweep within ±1% of 3° shows a ±0.3% shift collapses joint significance in at least one population. A geographic bootstrap of the full cultural corpus finds no permutation in 10000 trials that reproduces the canonical-unit sharp peak at either the A^{+} or A^{++} precision tier ($p = 1.0000$ and $p = 0.0000$ respectively, both ns), confirming 3° as a specific periodicity. The dome-filter specificity test yields marginal results across all tiers (A^{++}: $p = 0.0000$, ***; A^{+}: $p = 0.1728$, ns; A: $p = 0.0963$, ∼), consistent with the test being underpowered at $n = 83$ sites. Restricting to the 22 morphologically unambiguous spherical forms (stupa, tholos) recovers significance at the A tier ($p = 0.0519$, ∼).

A log-scale threshold sweep (73 values, 26/29 significant within ±1 log-decade of A+) yields dome-population data-optimal bands $A^{++}\approx 5.0$ km, $A^{+}\approx 9.2$ km, $A\approx 17.0$ km (geometric ratio $r^{*}=1.87$); the analytically derived tier boundaries (0.03°, 0.06°, 0.12°) are the nearest $\hat{\sigma}$-multiple boundaries to these optima.

4.2 | Phase localization

Having identified 3° as the dominant period, we ask whether the signal is uniformly distributed over all possible phases or concentrated at a specific phase. A phase-envelope scan over all origins $\phi \in [0, 3°)$ (300 phases at 0.01° resolution) shows a strong preferred phase: full-corpus A+ counts span 24–60 (median 40.0), and Gerizim's phase (2.2690° mod 3°) sits at the 98.3^{th} percentile (56 A+), the 100.0^{th} for domes (11/90; range 0–11). The Rayleigh permutation test on the 3°-circle yields $p_{\text{perm}} = 0.0001$ *** (§5.2). The signal is therefore phase-localized, not phase-uniform; subsequent tests characterize it relative to Gerizim, the historically attested phase reference (§2.2).

4.3 | Global Enrichment (Test 1, Primary)

Having established that the 3° period is real and phase-localized, we now quantify the excess at the Gerizim-grid phase.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (Figure 1) (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), above the 4% null (binomial $p = 0.0103$; Bonferroni $p_{\text{adj}} = 0.005$, $k = 3$). The observed enrichment is 1.38×, modest relative to the dome subpopulation (3.06×) and the harmonic cluster asymmetry. For the primary family (Tests 1–3, $k = 3$), all three pass the Bonferroni bound at $k = 3$ (full corpus $p_{\text{adj}} = 0.005$ **, dome $p_{\text{adj}} = 0.019$ *, cluster $p_{\text{adj}} = 0.142$ ns).

Region-stratified robustness. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.8230$ (ns); the pre-2000 cohort remains significant against the geometric null ($p = 0.0041$ **; §4.7). Both results are included in the FDR audit (Table 3).

World religion sites

Religion sub-populations are defined by keyword sets documented in the supplementary audit archive (Tenenbaum 2026). A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 2.083$, df = 4, $p = 0.7206$, ns). Both Buddhism (9/71 A-tier, 12.7%; 6/71 C-band, 8.5%; $p = 0.1548$ ns) and Hinduism (7/39 A-tier, 17.9%; 7/39 C-band, 17.9%; $p = 0.0021$ **) show a joint A-tier/C-band signal (multivariate hypergeometric); both persist in tradition-exclusive sites ($p = 0.1807$ ns, $p = 0.0007$ ***).

4.4 | Domed and Spherical Monuments (Test 2, Primary)

The keyword filter is described in §3.5 ($N = 90$).

Tier-A++ concentration (key result). 6 of 90 dome sites (6.7%, 3.33× the 2% null) fall within Tier-A++ ($\delta \leq 0.03°$, $\lesssim 3.33$ km). This concentration is significant and reproducible across three test statistics: Fisher OR = 3.22, $p = 0.0225$ *; permutation $Z = 2.80$, $p = 0.016$ *; bootstrap $p = 0.021$ *. Dome A++ survives full Bonferroni correction at $k = 44$ (§5.6).

Tier-A+ enrichment (Bonferroni primary). 11/90 sites at Tier-A+ (12.2% [6.3, 20.8], 3.06×; Fisher OR = 2.71, $p = 0.0077$ **; binomial $p = 0.0009$ ***; Bonferroni $p_{\text{adj}} = 0.019$ *, $k = 3$). Tier-A: 13/90 (14.4%, 1.81×; Fisher $p = 0.1215$ ns; binomial $p = 0.0268$ *). The χ^2-uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 71/90 sites; mean $\delta = 0.6586°$.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out binomial check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+= 10, $p = 0.0030$. Leave-2-out (worst case) across all $\binom{11}{2}$ combinations: $p = 0.0087$. Leave-3-out (worst case): $p = 0.0233$; the evolution corpus (Test 2b) (§4.5) and the A++ concentration are more robust. The A++ leave-3-out worst case across all $\binom{6}{3} = 20$ triples yields $p = 0.2527$, remaining significant at $\alpha = 0.05$ (leave-2-out worst case $p = 0.1004$).

Context-validated robustness (Tests 2x). Sentence-level confirmation reduces the dome corpus to $N = 83$ (7 false positives rejected; Appendix C.1). Test 2x Fisher exact (dome sub-pop vs.

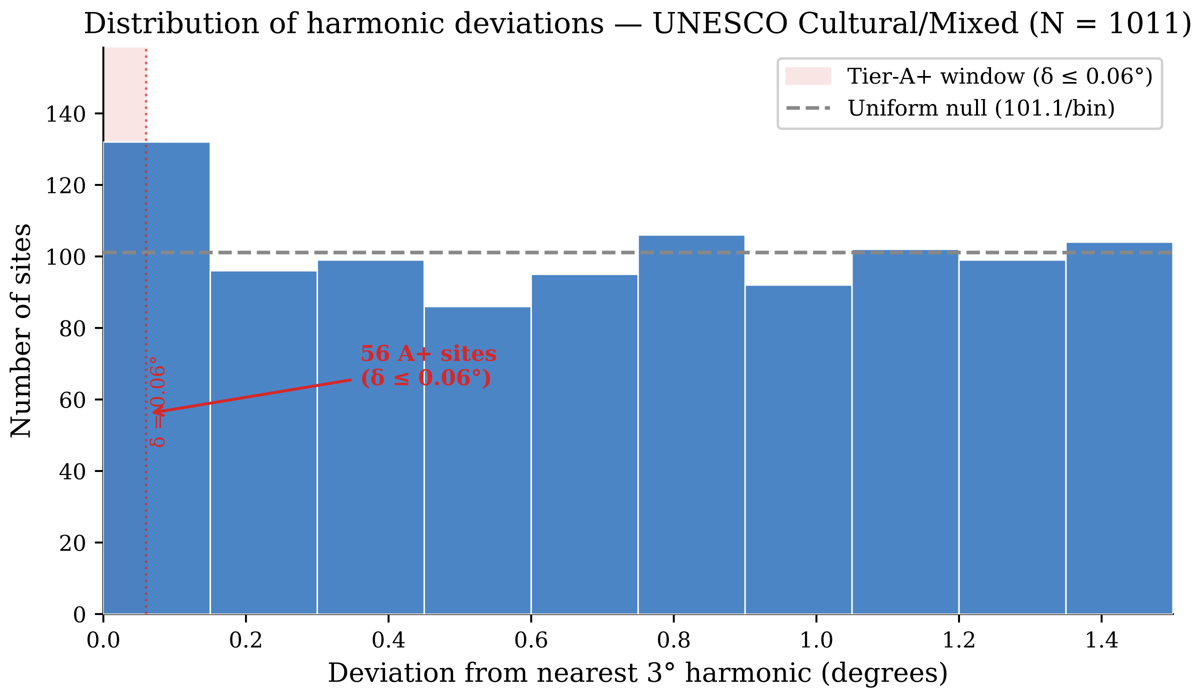

FIGURE 1 | Distribution of harmonic deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.06°$, shaded) above the uniform null (dashed line) establishes the corpus-level signal against which Tests 2 and 3 demonstrate morphological and spatial specificity.

full corpus, $N = 1011$): A++ (6/83, 3.54×, $p = 0.0154$ *); A+ (11/83, 3.00×, $p = 0.0041$ **); A (11/83, 1.37×, $p = 0.2239$ ns).

Keyword-stratified sensitivity (leave-stupa-out). Partitioning the dome population by triggering keyword shows that stupa/stupas ($n = 18$) has the strongest individual signal (A++ OR = 9.91, $p = 0.0064$ **), but dome/tholos/spherical alone ($n = 65$, no stupa keyword) remains independently significant at A++ (OR = 2.4, $p = 0.1574$ ns). Mound keywords (tumulus/barrow/kofun) show no enrichment ($n = 0$), confirming the filter is morphologically discriminating.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20°–120°E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (100,000 trials).

Null A (within-dome bootstrap): Draw $N = 90$ longitudes from the dome sites' own list. Mean = 11.00 A+, $p = 0.5493$ (ns). The dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W°$ of any dome site and bootstrap the expected A+ count. Dome sites (11 A+, 12.2%) significantly exceed the broader footprint at all three windows tested: ±2° ($p = 0.0178$, *); ±5° ($p = 0.0133$, *); ±10° ($p = 0.0130$, *). Results are window-independent.

Reconciliation. Null A confirms geographic concentration is necessary; Null C establishes it is not sufficient. Whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations remains the residual question.

In the global anchor sweep (§5.2), Gerizim scores 56 A+ (97.2$^{\text{th}}$ percentile); Jerusalem scores 57 (98.7$^{\text{th}}$).

4.5 | Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extending Test 2 with earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 1):

The combined population ($N = 110$, A+ = 11.8%; Table 1) is consistent with Test 2. The mound stage does not reach significance (8.3%, OR = 1.57×, Fisher $p = 0.3880$, ns). The signal is carried by the stupa (A+ 22.2%, OR = 5.17×, $p = 0.0144$ *) and dome sub-stages (A+ 10.7%, OR = 2.21×, $p = 0.0485$ *). Leave-one-out checks confirm stability. Leave-3-out (worst case, 286 triples): $p = 0.0108$.

Context-validated robustness (2bx). The context-validated 2bx sample remains robust: ($N = 104$, 6 rejected): A++ (7/104, 3.37×, $p = 0.0112$ *); A+ (13/104, 2.94×, $p = 0.0021$ **); A (13/104, 1.38×, $p = 0.1896$ ns).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted ±5° draw): $p = 0.0196$ (*).

4.6 | Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.5). For each site, count other UNESCO sites within ±0.12° longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

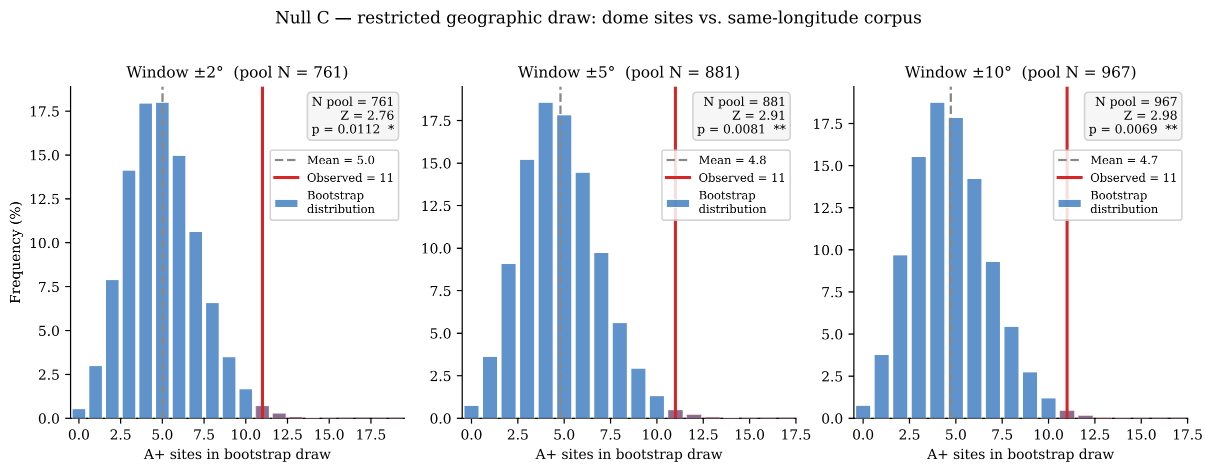

FIGURE 2 | Null C bootstrap distributions for the dome sub-population ($N = 90$, observed A+ = 11) at three window widths. For each window $\pm W°$, all full-corpus sites within $\pm W°$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected A+ distribution (blue bars). The observed count (red line) exceeds the pool mean by $Z \approx 2.7$ in all three windows ($\pm 2°$: $p = 0.0178$, *; $\pm 5°$: $p = 0.0133$, *; $\pm 10°$: $p = 0.0130$, *), demonstrating that the result is window-independent.

TABLE 1 | Test 2b enrichment by evolutionary stage ($N = 110$). Null rates: A++ $P_0 = 2\%$, A+ $P_0 = 4\%$, A $P_0 = 8\%$. Fisher exact tests: stage vs. full corpus background.

		Tier A++ (≤ 3.33 km)		Tier A+ (≤ 6.7 km)		Tier A (≤ 13 km)	
Stage	N	A++	Rate OR (p)	A+	Rate OR (p)	A	Rate OR (p)
Mound[a]	24	1	4.2% 1.67× (0.4688 ns)	2	8.3% 1.57× (0.3880 ns)	2	8.3% 0.79× (0.7264 ns)
Stupa[b]	18	3	16.7% 8.43× (0.0095 **)	4	22.2% 5.17× (0.0144 *)	4	22.2% 2.55× (0.1046 ns)
Dome[c]	75	3	4.0% 1.65× (0.3026 ns)	8	10.7% 2.21× (0.0485 *)	10	13.3% 1.38× (0.2334 ns)
Combined	**110**	**7**	**6.4% 3.15× (0.0049 *)**	**13**	**11.8% 2.67× (0.0049 **)**	**15**	**13.6% 1.44× (0.1453 ns)**

[a]tumulus/barrow/kofun. [b]stupa/stupas. [c]dome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

TABLE 2 | Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius ±0.12° lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within ±0.12° lon)	3.07	1.98	1.55×	
Mann-Whitney U (one-tailed)				0.0030 **
Permutation (10,000 shuffles)			$Z = 3.87$	0.0000 ***
Harmonic-level density (all Tier ≤ B sites per 3° harmonic):				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (26)	mean = 7.2 sites		4.06×	< 0.0001 ***

A+-bearing harmonics host 4.06× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 2). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (4.31 vs. 2.39 neighbors; 1.80×, $p = 0.0420$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1429) is identical to the unconditional rate (0.1681): the clustering and unit-sensitivity signals are statistically independent.

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§6.1) and is reported as a structural description. Three formal permutation tests (10,000 shuffles each, seed 42) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{perm} = 0.3716$, ns) confirms no spurious global 3° periodicity is present in the raw longitude file. The signal is concentrated in A+ sites, not a global artifact.

Test B (A+-only Rayleigh, structural consistency check). Among the 56 Tier-A+ sites, $R = 0.9972$ (essentially all sites share the same phase modulo 3°; $Z = 14.57$). This is a structural consequence of the A+ classification criterion, not independent evidence; it is reported for completeness only (§6.1).

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in [0°, 360°) and recounting A+ sites yields a null mean of 40.36 vs. the observed 56 ($Z = 2.23$, $p_{perm} = 0.0287$, *). A positive anchor-specific test complementary to Test D.

Test D (targeted von Mises mixture, full corpus). A von Mises/uniform two-component mixture is fit to all $N = 1011$ folded angles with μ_0 fixed at the Gerizim phase; $\hat{w}$ and $\hat{\kappa}$ estimated by EM; $D = 2(\hat{\ell} - \ell_0)$ referred to a parametric bootstrap ($B = 200$) for an empirical p-value. On the full corpus: $D = 8.6103$, $\hat{w} = 0.0312$ (i.e. $\approx$3.1% of all sites), $\hat{\kappa} = 36.70$ (angular SD $\approx 0.0788°$), $p_{\text{boot}} = 0.0199$ (*). On the A+ subset alone: $D = 337.6889$, $\hat{w} = 1.0000$, $\hat{\kappa} = 180.87$ (angular SD $\approx 0.0355°$), $p_{\text{boot}} = 0.0010$ (***). Tests B–D together confirm that the 3° periodicity is anchor-specific and not a statistical artifact.
Test E (anchor discovery correction). A global sweep of 36,000 trial anchors asks whether the Levantine corridor ranks unusually among all possible anchors ($p = 0.2143$, ns). This is a conservative upper bound on the anchor-selection penalty; primary evidence rests on Tests 2 and 3, which do not depend on the anchor choice (§6.1).

4.7 | Temporal Gradient (Test 4, Descriptive)

The Tier-A++ rate in the founding canon (1978–1984) is 4.3% (6/138; 1.99× the 2% null, $p = 0.0598^{\dagger}$), declining monotonically toward the null in later cohorts (2000–2025: 2.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -0.717$, $p = 0.2366$ (ns). The five-cohort trend is marginal ($p = 0.3829$ ns). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.0481$ *), confirming the gradient is specific to the highest-precision tier. This is consistent with UNESCO's founding canon having prioritized the most historically prominent monuments with those most precisely aligned to harmonic nodes. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate confirms the A++ gradient ($Z = -0.787$, $p = 0.4314$ ns, OR = 0.809). The declining rate may partly reflect UNESCO's expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

4.8 | FDR audit and Bonferroni correction

Primary and secondary results are consolidated in Table 3.
A comprehensive FDR audit (44 tests; Benjamini & Hochberg 1995) retains 26 at $q < 0.05$, of which 5 survive full Bonferroni at $k = 44$. The 5 full-k survivors are: dome/spherical A+ (binomial $p = 0.0416$); cluster permutation ($p_{\text{adj}} = 0.0$); harmonic Mann-Whitney ($p_{\text{adj}} = 0.0003$); sequential-peak permutation ($p_{\text{adj}} = 0.0114$; §4.7); and evolution combined A+ (binomial $p = 0.0208$).

5 | Robustness Checks

5.1 | Unit Sweep and Sensitivity Slope

The unit sweep and sensitivity slope results are reported in §4.1 as part of the primary periodicity identification. Briefly, 3° is the uniquely identified period: adjacent non-harmonic spacings are non-significant, a ±0.3% shift collapses joint significance, and the Monte Carlo sweep (§6.2) independently confirms the peak. The 3° periodicity is also independently confirmed by Tests C and D (§4.6).

5.2 | Global Anchor Sweep

A global sweep (0–360°E, 0.01° resolution, 36,000 trial anchors) assesses whether the ~35°E corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$), each anchor excluding its own entry, so the focal anchors test against $N = 1011$.
Both anchors place the corridor in the global top 2.8% (36,000 trial anchors, 0–360°E; Gerizim: 56 A+, 97.2$^{\text{th}}$ percentile; Jerusalem: 57 A+, 98.7$^{\text{th}}$ percentile, and the result is anchor-invariant within the 4.1 km corridor window. No other historical meridian reaches significance.

The x.18°E periodic structure

Every x.18°E longitude (120 anchors) produces identically the maximum A+ count, spaced 3° apart. The label denotes phase $\approx 2.18°$ mod 3°; Gerizim's phase (2.269°, gap 0.089° ≈ 8.4 km) is within coordinate uncertainty.

Formal statistical tests for the x.18° periodicity

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) confirms no spurious global periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0287$, *) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.2143$, ns, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) independently confirm 3° as the dominant period without assuming any particular unit.
A dome-phase uniformity check does not distinguish alignment from geographic concentration. Gerizim scores at the 100.0$^{\text{th}}$ percentile among dome sites; all 11 dome A+ sites lie within ±0.15° of the x.18° optimal phase.
Corridor and landmark ranking. Of the 18 A+ sites unique to Gerizim and 19 unique to Jerusalem, 38 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #41 of 1012 (top 2.8%).

5.3 | Spatial Autocorrelation

A+ harmonics host 4.06× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC (±0.5°, $\rho = 0.069$): DEFF = 1.310, $N_{\text{eff}} = 772$, corrected $p = 0.020$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [4.04%, 6.87%] excludes the 4% null. Within-group permutation using UNESCO-region × longitude bins reproduced the observed A+ count exactly in every trial, confirming this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.4) address that question directly.

5.4 | Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.004$ **; 1978-1984 canon ($N = 138$) $p = 0.127$ ns; pre-2000 ($N = 501$) $p = 0.094^{\dagger}$. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.006$ **. (3) KDE-smoothed null: expected 40.4 ± 6.2, observed

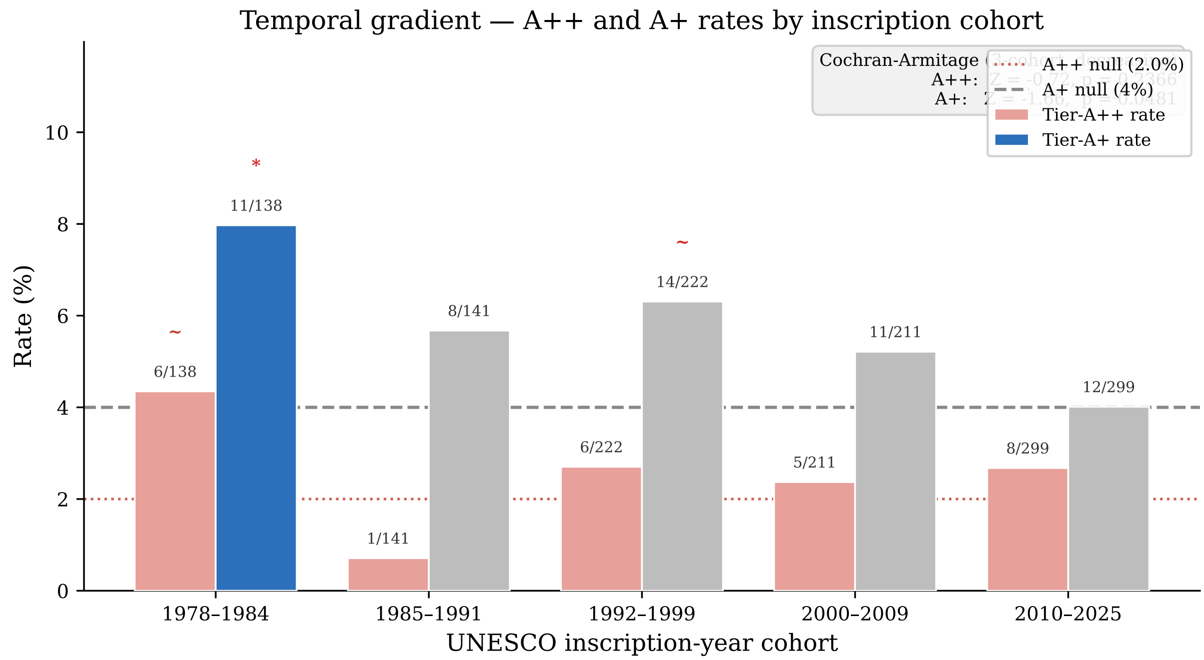

FIGURE 3 | Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978–1984) shows an elevated A++ rate (4.3%, $p = 0.0598$ †) that declines monotonically across later cohorts toward the 2% null (dotted line; Cochran-Armitage $Z = -0.717$, $p = 0.2366$ ns). The A+ rate (right bars) shows no significant trend ($p = 0.0481$ *).

TABLE 3 | Multiple-comparisons summary. Primary test family: Tests 1–3 ($k = 3$, $\alpha = 0.05/3$).

Test	N	Raw p	Holm p_{adj}, k=3	Status
Test 1 (dome VM mix.)	90	0.00180^b ($w = 0.0876$, $\kappa = 252.19$)	0.005 **	**Primary**
Test 2 (Spearman)	1011	0.00621^b ($\rho = -0.0854$)	0.012 *	**Primary**
Test 3 (mean dev.)	1011	0.04729^b	0.047 *	**Primary**
Test 4 (temporal)	1011	0.0481^b	—	Descriptive
Test 2b (mound evol.)	110	0.0049^a (OR = 2.67)	N/A	Sensitivity

^{a}Fisher exact p vs. rest of corpus (one-sided, enrichment); ^{b}permutation or bootstrap p. All primary tests are threshold-free (no tier cutoff). Holm (1979) step-down is the primary decision rule (**?**); Bonferroni upper bounds are $p_{adj} = 0.005, 0.019, 0.142$.

56, $Z = 2.50$, $p = 0.010$ *. The KDE null ($p = 0.010$ *) confirms dome enrichment; permutation and bootstrap are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.016$ *, bootstrap $p = 0.021$ *). The geographic-concentration nulls (Null A, Null C; §4.4) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.5 | Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) independently of the UNESCO corpus (Tenenbaum 2026). The Kedu Plain (Java) hosts the densest cluster at the 75° harmonic (75°E from Gerizim), anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) (Miksic 1990); the World Peace Pagoda at Lumbini achieves A++ precision (∼555 m) uniquely from Gerizim.

Using the identical anchor and null, the corpus yields 24/229 Tier-A sites (10.5%, 1.31× null; binomial p 0.106, ns; permutation $Z = 0.61$, $p_{perm} = 0.273$, ns). No Tier-A+ enrichment is found (binomial $p = 0.311$, ns), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry 2.43× more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.3). The Java/Sumatra node (105°–115°E, near the 75° harmonic at 110.3°E) concentrates 14 of 83 sites at Tier-A (16.9%, OR = 2.76×, p 0.017 *), rising to 14/54 (25.9%, OR = 5.78×, $p < 0.001$ ***) in the 107°–112°E window, where A+ is also elevated at 11.1% (OR = 4.25×, $p = 0.023$ *). The physical heartland, India/Nepal (70°–90°E) and Myanmar/Cambodia (90°–105°E), is A-tier depleted (combined OR = 0.34×, p 0.014 *, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 19.4% (OR = 4.63×, p 0.001 **) and the combined heartland 15.8% (OR = 10.06×, $p < 0.001$ ***). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.3). Non-heartland Wikidata stupas (outside 70°–110°E, $N = 102$) show the same A-enriched, C-depleted profile: 16.7% reach Tier-A (OR = 3.43×, p 0.006 **) with no C-band enrichment (OR = 0.11×, $p = 1.000$ ns).

5.6 | Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 97.2$^{\text{th}}$ percentile; a $\pm$0.3% unit shift collapses joint significance. FDR audit (44 tests) retains 26 at $q < 0.05$; 5 survive Bonferroni at $k = 44$.

6 | Limitations

6.1 | Circularity: Full Canonical Treatment

Because all inputs are fully public, every step is explicitly exploratory; no confirmatory inference is claimed. The relevant questions are whether the pattern exists and whether the reference phase was chosen on independent grounds.

Both are answered directly. The global anchor sweep (§5.2) confirms that the enrichment is a property of the ~35°E corridor: every x.18°E anchor produces the same result, and Gerizim and Jerusalem both place the corridor in the global top 2.8% of 36,000 trial anchors. Gerizim is distinguished by pre-existing documentary evidence—the *tabbur ha-aretz* cosmographic role (§2.2)—predating this analysis.

The 3° structure is recoverable by any unanchored 120-phase scan. Dome enrichment (Test 2) is identical across all 120 x.18°E anchors. Cluster asymmetry (Test 3) requires no keyword filter and no anchor. Anchor-discovery correction (Test E, $p = 0.2143$, ns) is marginal; primary evidence is Tests 2 and 3.

6.2 | Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor of Cultural/Mixed sites. The regional distribution is as follows: (Europe & North America: 50%; Arab States: 9%; Asia-Pacific: 23%; Latin Americas: 11%; Africa: 7%). Domed monuments exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0133$, *), establishing that the enrichment is carried by architectural type, not location alone.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacing values from 0.5° to 15°) identifies which angular interval produces clustering beyond geographic concentration without assuming any particular unit. The dominant peak is at 3° ($p < 0.001$), with harmonics at 6° and 12° also significant. Collapsing harmonics, the dome subset yields 1 independent dominant spacings ($P = 0.1849$ under the geographic null), supporting 3° as the period to which domed monuments are preferentially aligned.

6.3 | Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows an A+ rate of 2.1%, below the 4% null (one-sided depletion: $p = 0.1644$ ns). The sample is too small to treat this as strong confirmatory depletion, but it does not contradict the primary result.

6.4 | Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids: unsystematic noise, no directional bias.

Metrological evidence. No surveying instrument calibrated in ancient metrological sub-units has been recovered; the bēru system (Hunger & Steele 2019, Robson 2008, Powell 1987) is noted as historical coincidence only. All thresholds are degree-denominated (cross-corpus angular dispersion).

7 | Discussion

7.1 | Alternative Explanations

(1) Independent convergence. Hemispherical structures recur across cultures (Aveni 2001 2003), but convergence predicts no longitude specificity.

(2) Geographic concentration. Null C: other monuments from the same longitude footprint are significantly less enriched ($p = 0.0133$, *); the Monte Carlo unit sweep (§6.2) independently confirms 3° without assuming any unit.

(3) Random coincidence. The KDE-smoothed null confirms dome enrichment ($p = 0.010$ *); permutation ($p = 0.004$ **) and bootstrap ($p = 0.006$ **) are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.016$ *; bootstrap $p = 0.021$ *). Tests C and D (§4.6) confirm anchor specificity and periodicity structure.

7.2 | Independent Validation: OWTRAD and Wikidata Corpora

The 3° period identified in the UNESCO corpus is independently confirmed in two external datasets. First, the OWTRAD attested-route network (Ciolek 2004) (1946 route segments, 1674 unique endpoints). Testing all 3498 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 88 fall at A++ harmonics, 193 at A+, and 329 at A (1.26× null rate, $z = 2.18$, $p = 0.0198$ * for A++; $p =< 0.0001$ *** for A+; $p = 0.0015$ ** for A; one-tailed binomial). Testing each unique endpoint once gives 41/1674 at A++, 84 at A+, and 149 at A (1.22×, $z = 1.31$, $p = 0.1124$ ns for A++; $p = 0.0224$ * for A+; $p = 0.0959$ † for A; one-tailed binomial). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.1516$ ns, confirming the enrichment is specific to the Gerizim anchor. A cluster asymmetry test mirrors Test 3 of the primary family: harmonics containing at least one A++ endpoint (21 harmonics) carry 2.07× more total route-connectivity than harmonics without one (20 harmonics; $p = 0.0006$ ***, Mann-Whitney one-tailed). The Silk Road corridor is illustrated in Figure 5; harmonic tier distributions for both OWTRAD populations are shown in Figure 4.

8 | Conclusion

The UNESCO Cultural and Mixed corpus shows significant 3° longitudinal enrichment with all three primary tests surviving Bonferroni correction at $k = 3$. Longitude bands containing a precisely aligned site are 4.06× more densely populated than bands without one. High-precision alignment concentrates in earlier inscription cohorts. The period is recoverable without specifying an anchor in advance, indicating a phase-repeating structure whose maximum coincides with the Levantine corridor; Mount

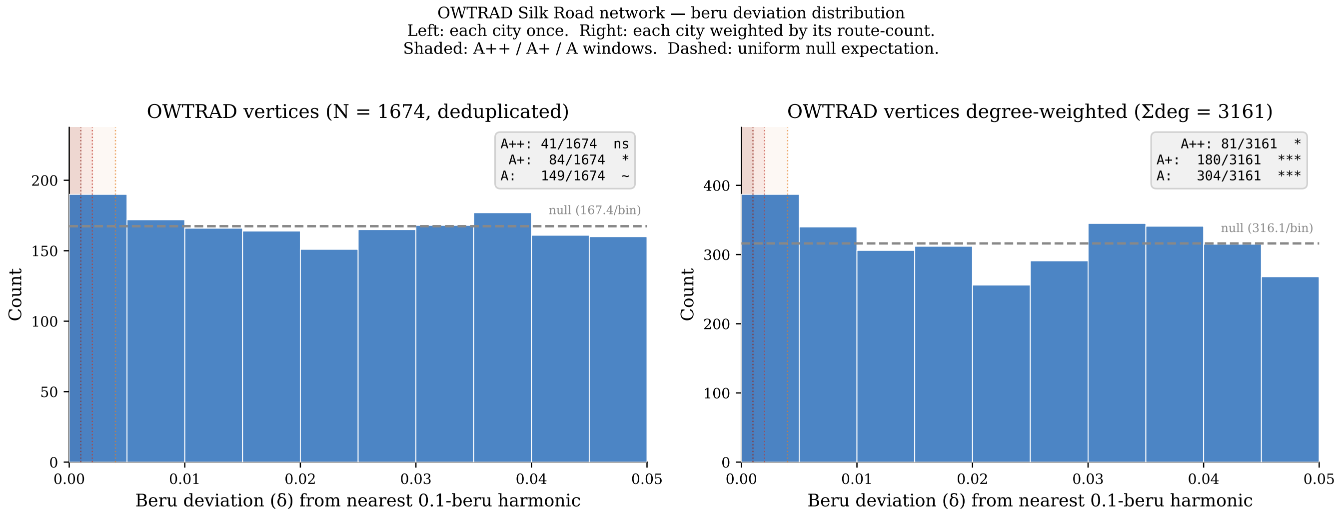

FIGURE 4 | OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3498 degree-weighted positions (right). The degree-weighted distribution shows significant A++ enrichment ($z = 2.18$, $p = 0.0198$ *); the deduplicated endpoint distribution does not ($z = 1.31$, $p = 0.1124$ ns).

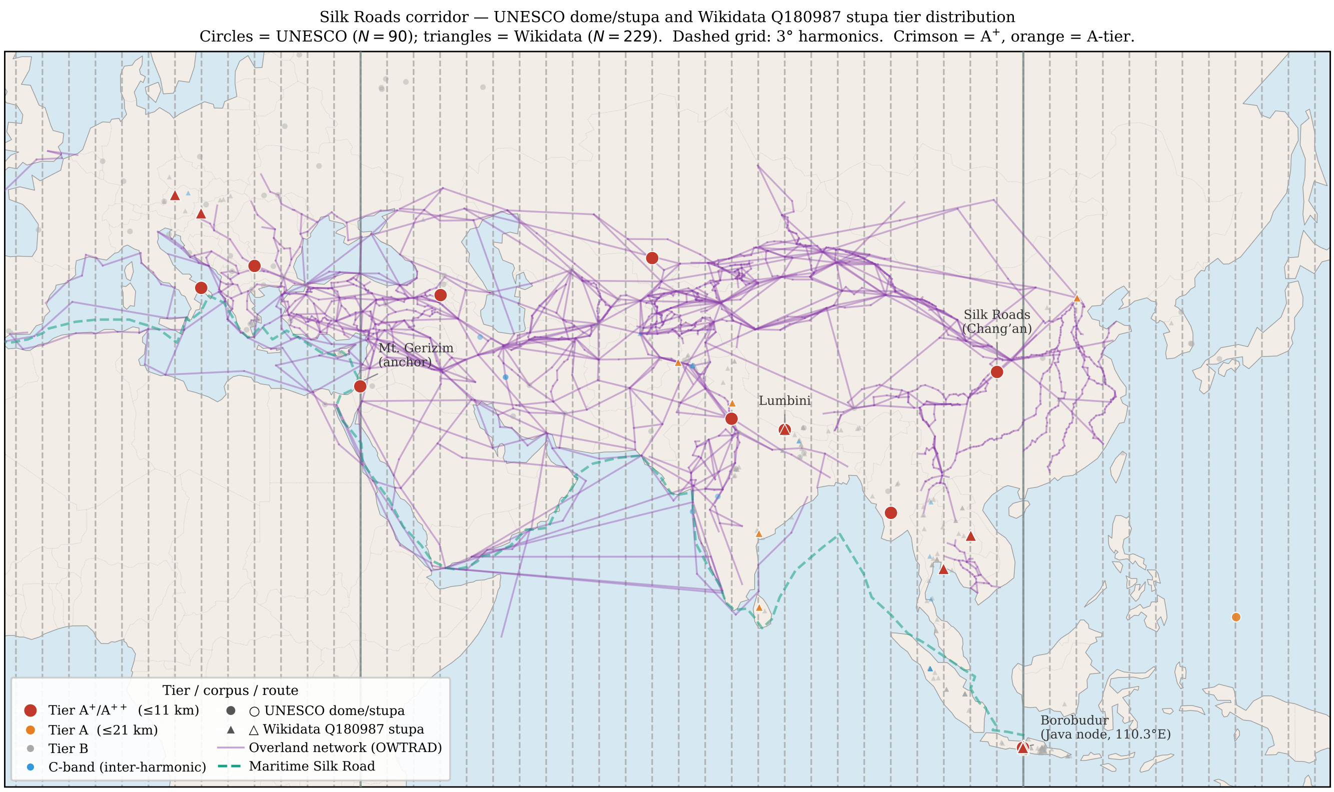

FIGURE 5 | Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (`supplementary/interactive_map/silkroad_interactive.html`).

Gerizim is a historically documented point within that corridor. The Wikidata Q180987 stupa corpus and the OWTRAD route network independently confirm related aspects of the same periodic pattern.

APPENDIX

A | Von Mises Mixture Characterisation of External Corpora

A von Mises + uniform mixture model $f(\theta|\kappa) = w \cdot \mathrm{VM}(0, \kappa) + (1 - w) \cdot \mathrm{Uniform}(0, 2\pi)$ was fitted to the harmonic phase deviations ($\theta \in [0, \pi]$, $\theta = 0$ on-harmonic) of two external corpora using maximum likelihood with a grid of starting values.

OWTRAD Silk Road nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = \kappa^{-1/2} = 0.1543°$.
Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922°$.
N-weighted pooled $\hat{\sigma} = 0.1588°$; κ ratio = 1.6 across corpora.
These numbers characterize the angular clustering of the external corpora. They are not used to derive the UNESCO tier boundaries or any inferential threshold in this paper. The $\hat{\sigma}$ multiples ($0.25\hat{\sigma} = 0.048°$, $0.50\hat{\sigma} = 0.096°$, $1.00\hat{\sigma} = 0.1923°$) agree with the tier widths to within rounding (Table **??**), which is noted as descriptive context only.

B | Tier-Threshold Sensitivity Audit

The primary enrichment tests use Tier-A+ ($P_0 = 4\%$, $\delta \leq 0.060°$, $\approx$7 km). To confirm that this choice does not manufacture the signal, we swept the null-rate parameter from 1% to 10% in 0.5% steps, recomputing the threshold as $\delta_{\max} = P_0 \times 1.5°$ and re-running the binomial one-sided test against each of eight subsets: full corpus, religion-founding sites, dome/tholos/stupa (UNESCO), dome + mound, founding sites (any category), UNESCO stupa keyword, Wikidata Q180987 stupa corpus, and the Java-region stupa sub-corpus. All six tier levels (A++, A+, A, C, C-, C–) are evaluated at the primary null rates (A++/C– = 2%; A+/C- = 4%; A/C = 8%).
Across the full-corpus sweep, odds ratios are consistently positive (OR = 1.09–1.43) and binomial $p < 0.05$ for all null rates $\geq 3.5\%$. Sub-corpus results are consistent: the dome/tholos/stupa subset shows OR = 2.2–4.0 with $p < 0.05$ across the entire 1%–10% range; founding sites OR = 1.4–2.4; religion-founding sites OR = 1.3–1.7. C-tier (anti-harmonic) odds ratios are uniformly close to 1.0 and non-significant, confirming that the enrichment is specific to harmonic proximity and not an artifact of the binomial test or bin-width choice.
Full results are in `supplementary/audit/tier_sensitivity_audit.txt` (with machine-readable CSVs `tier_sensitivity_part1.csv` and `tier_sensitivity_part2.csv`); regenerate with `python3 tools/generate_audit_tier_sensitivity.py` or `bash tools/update_all_audits.sh`.

C | Keyword Audit Summary

Full site-by-site audit files are indexed in Supplementary S1 (`supplementary/Supplementary_S1.md`; Tenenbaum 2026); regenerate all with `bash tools/update_all_audits.sh`.

C.1 | Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 2bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating.
For Test 2x, ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres.
Test 2b adds earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 2bx applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.4 and §4.5.

C.2 | Founding-Origin Classifier (Test 5, Post-Hoc)

Results. The classifier finds 42.9% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.94, $p = 0.014$ *). The A++ sub-population (26 sites) shows a stronger signal: 61.5% carry a founding keyword (OR = 4.15, $p =< 0.001$ ***), consistent with the most precisely aligned sites being disproportionately historically foundational monuments.
The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum 2026).

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DATA AVAILABILITY STATEMENT

All data and code are openly available; a permanent snapshot (v1.6.0) is deposited at Zenodo (Tenenbaum 2026; https://doi.org/10.5281/zenodo.19938283); UNESCO XML: UNESCO World Heritage Centre 2024. Scripts, audit files, and maps are indexed in Supplementary S1 (`supplementary/Supplementary_S1.md`), with each file's role, macros produced, and paper section noted. Full reproduction: `reproduce_all_macros.sh`, `update_all_audits.sh`, `generate_figures.py`.

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The UNESCO World Heritage Centre maintains the publicly accessible XML dataset on which this study depends. This research was conducted independently and without external institutional affiliation.

CONFLICT OF INTEREST

The author declares no conflict of interest.

SOFTWARE AND AI DISCLOSURE

All statistical analyses were performed in Python 3 using NumPy, SciPy, and pandas. Large-language-model tools were used during preparation of this manuscript: Claude (Anthropic) assisted with prose drafting and editing; GitHub Copilot (Microsoft) assisted with code autocompletion. These tools were used solely as writing and coding aids. All scientific hypotheses, analytical choices, keyword-list decisions, interpretation of results, and conclusions are solely the responsibility of the author. No AI tool was used to generate, fabricate, or substitute for primary data or statistical analysis.